

Perform Website, Email, Whois, DNS, and Network Footprinting on the Target Network

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Gather domain registration, DNS, mail, web technology, and network-path information for the synthetic lab target.

Theory

Whois and DNS records describe how names map to services. Traceroute shows the network path from the test workstation. The outputs below use a private host-only range and a documentation domain; no public DNS zone or real registrant was queried.

Tools and Lab Scope

- whois
- nslookup
- dig
- traceroute
- WhatWeb
- Netcraft/BuiltWith-style lab record
- Browser developer tools

Procedure and Recorded Execution

Step 1. Collect registration information - Query the local training record for the domain owner and name servers.

```
whois lab.example.test
Registrar: TRAINING-REGISTRAR
Status: clientTransferProhibited
Name Server: ns1.lab.example.test
```

Step 2. Enumerate DNS records - Record A, MX, NS, and TXT data. Do not attempt a zone transfer outside an authorized lab.

```
dig lab.example.test A MX NS TXT
A 192.168.56.30
MX 10 mail.lab.example.test.
NS ns1.lab.example.test.
```

Step 3. Fingerprint the website - Review page source and a local technology fingerprint.

```
whatweb http://lab.example.test
Apache[2.4.57], PHP[8.2], HTML5
```

Step 4. Trace the path - Use the host-only route to confirm that all hops are inside the lab.

```
traceroute lab.example.test
1 192.168.56.1 0.4 ms
2 192.168.56.30 0.6 ms
```

Step 5. Review email headers - Use a synthetic test message and redact message IDs before submission.

Consolidated footprint output

All observed infrastructure details were consistent with the isolated lab network. The DNS record set exposed no real organization data.

Record/source	Observed value
Whois	TRAINING-REGISTRAR; created 2025-01-12; privacy-protected registrant
A record	lab.example.test -> 192.168.56.30
MX record	10 mail.lab.example.test -> 192.168.56.25
NS record	ns1.lab.example.test -> 192.168.56.2
TXT	v=spf1 -all (training record)
Web stack	Apache 2.4.57; PHP 8.2; HTML5
Traceroute	192.168.56.1 -> 192.168.56.30; 2 hops

Result

Website, email, Whois, DNS, and network footprinting completed. The result is a compact infrastructure map suitable for planning later defensive testing.

Safety and Submission Notes

- Do not query real domains without written authorization.
- Treat email headers and registrant data as sensitive.
- Zone-transfer checks belong only in a lab or explicitly approved assessment.

Viva Questions and Answers

What does Whois provide?

Registrar, registration dates, name servers, status, and sometimes registrant information.

A vs MX record?

An A record maps a name to an IP address; an MX record identifies mail servers and their priority.

What is a DNS zone transfer?

A replication operation that can disclose an entire zone if a server permits unauthorized transfers.

How can page source help?

Comments, library versions, hidden fields, and endpoint names may reveal implementation details.

What does traceroute show?

The sequence of routing hops between the test host and destination, subject to filtering and load balancing.